Day 2: DHCP Configuration

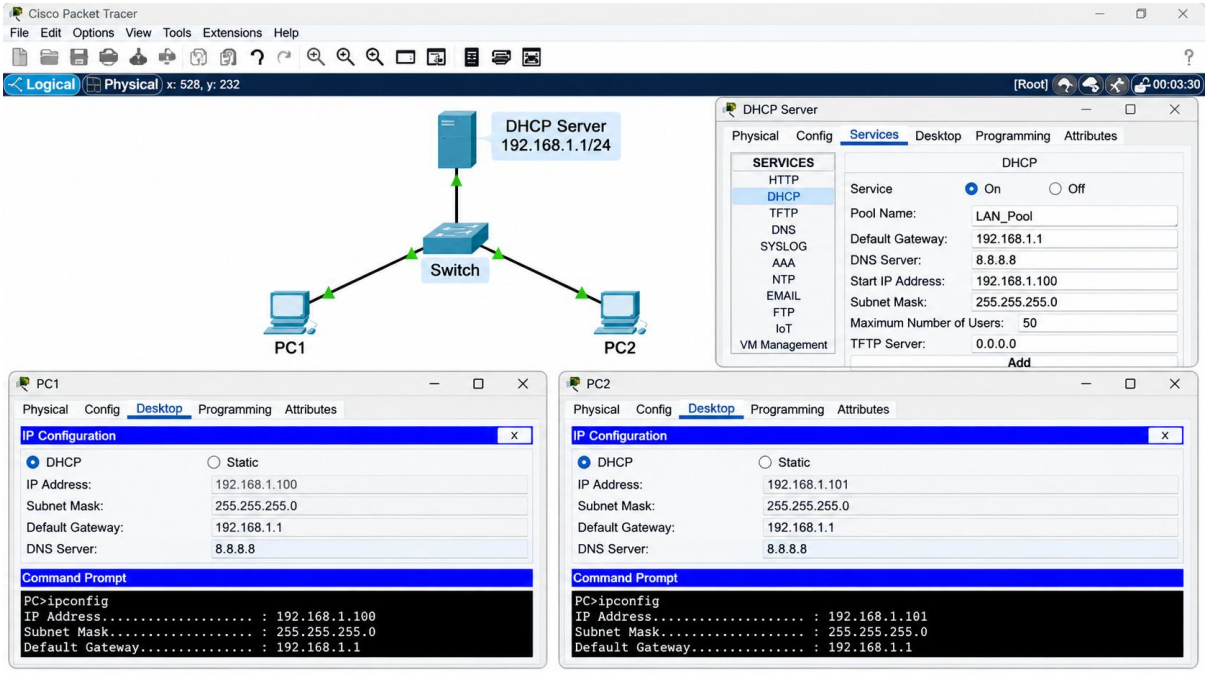
DHCP process (Discover, Offer, Request, Acknowledge):

Step	Message	From > To	Purpose
1	DHCP Discover	Client > Server (broadcast)	The client has no IP address, so it asks "Is there a DHCP server?".
2	DHCP Offer	Server > Client	The server offers an IP address from its pool, with subnet mask, gateway, DNS and lease time.
3	DHCP Request	Client > Server (broadcast)	The client accepts the offered address and asks for it.
4	DHCP Acknowledge	Server > Client	The server confirms and leases the address. The client can now use it.

Task 1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Answer:

Devices: 1 switch (2960), 1 server (Server-PT, named DHCP Server) as the DHCP server, and 2 PCs (PC1 and PC2). All are connected with copper straight-through cables (server and PCs to the switch).



Screenshot: DHCP server (LAN_Pool, start IP 192.168.1.100), PC1 and PC2 in Cisco Packet Tracer, with the IP configuration and ipconfig output of each PC

Device	Connected to	Role	Address
DHCP Server	Switch	DHCP server	192.168.1.1/24 (static)
PC1	Switch	DHCP client	From DHCP
PC2	Switch	DHCP client	From DHCP

Steps:

1. Place the devices in Packet Tracer and connect the server, PC1 and PC2 to the switch with straight-through cables. Wait until the link lights turn green.
2. Give the server a static IP address: click Server > Desktop > IP Configuration > Static. Enter IP 192.168.1.1, subnet mask 255.255.255.0.
3. Open Server > Services > DHCP and turn the Service On.
4. Fill in the pool: Pool Name = LAN_POOL, Default Gateway = 192.168.1.1, DNS Server = 8.8.8.8, Start IP Address = 192.168.1.100, Subnet Mask = 255.255.255.0, Maximum Number of Users = 50. Click Save (or Add).
5. On PC1: Desktop > IP Configuration > select DHCP. Wait a few seconds and note the IP address, mask, gateway and DNS received. Repeat on PC2.
6. Verify in the Command Prompt of each PC with the command ipconfig /all.

Configuration used on the DHCP server:

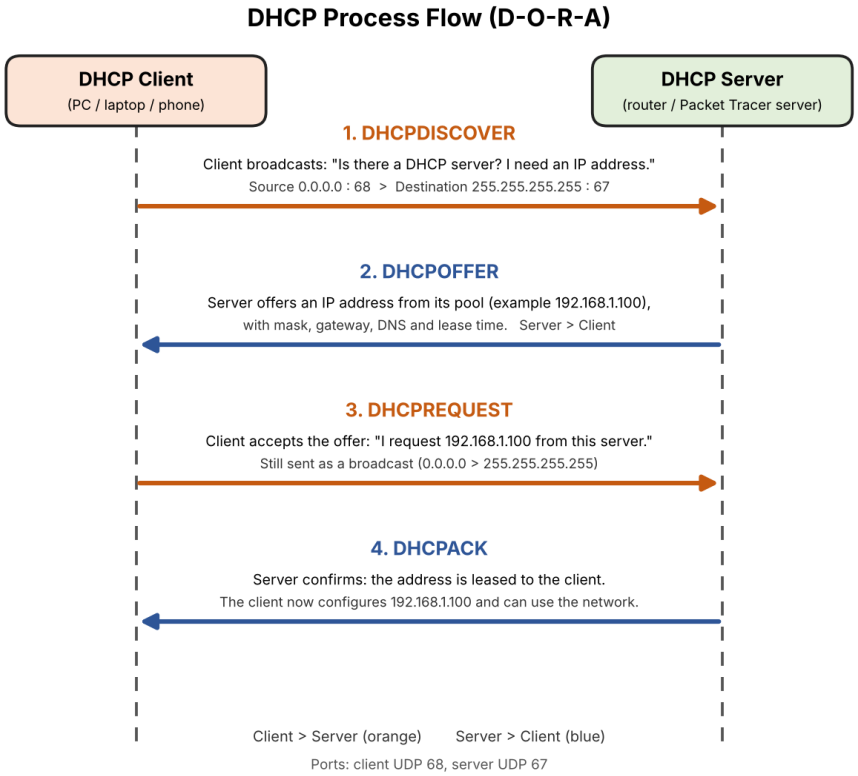
Setting	Value
Server static IP / mask	192.168.1.1 / 255.255.255.0
Pool name	LAN_POOL
Start IP address	192.168.1.100
Maximum number of users	50 (so the pool is 192.168.1.100 to 192.168.1.149)
Default gateway / DNS	192.168.1.1 / 8.8.8.8

Result: DHCP gave each PC a different address from the pool: PC1 received 192.168.1.100 and PC2 received 192.168.1.101. Both also received the subnet mask 255.255.255.0, the default gateway 192.168.1.1 and the DNS server 8.8.8.8, which is exactly what is configured in LAN_Pool. The server's own address 192.168.1.1 is outside the pool, so it can never be given to a PC.

Device	IP address received	Subnet mask / Gateway / DNS
PC1	192.168.1.100	255.255.255.0 / 192.168.1.1 / 8.8.8.8
PC2	192.168.1.101	255.255.255.0 / 192.168.1.1 / 8.8.8.8

Task 2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Answer:



DHCP sequence diagram: Discover, Offer, Request and Acknowledge

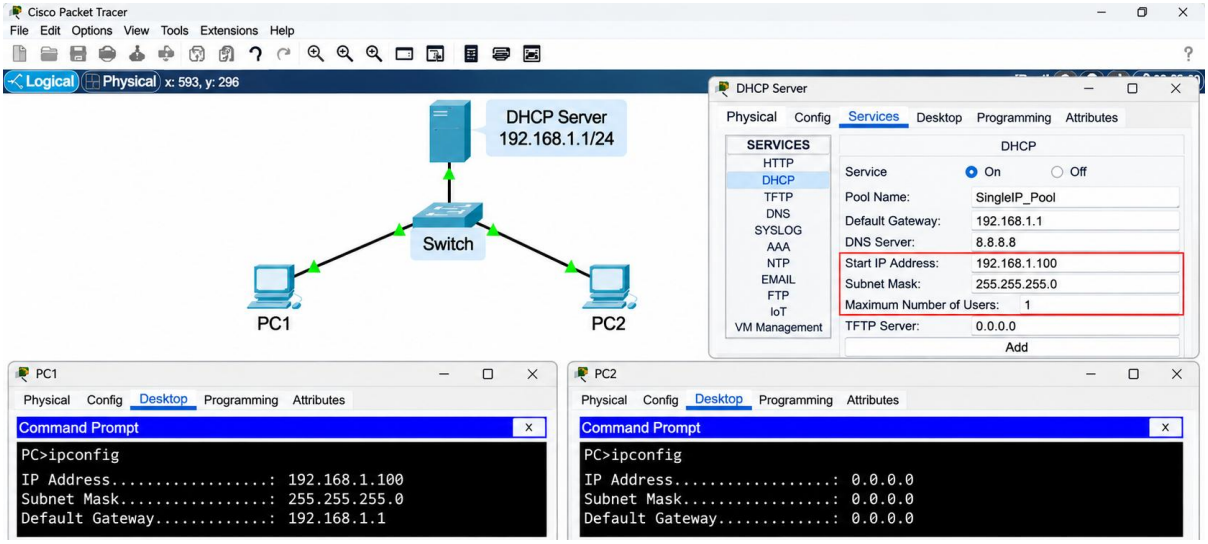
Explanation: Orange arrows go from the client to the server and blue arrows go from the server to the client. The Discover and Request messages are broadcasts (destination 255.255.255.255), because the client does not have a valid IP address yet. The client uses UDP port 68 and the server uses UDP port 67.

Task 3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Answer:

1. Use the same network as Task 1 (server, switch, PC1 and PC2).
2. Open Server > Services > DHCP. Create a pool named SingleIP_Pool (or edit the pool), keep Start IP Address = 192.168.1.100 and change "Maximum Number of Users" from 50 to 1. Click Save. Now the pool has only one address (192.168.1.100).
3. On PC1: Desktop > IP Configuration > DHCP. Note the result.
4. On PC2: Desktop > IP Configuration > DHCP. Note the result.

Device	Result (from my screenshot)
PC1 (asks first)	Gets 192.168.1.100 (with gateway 192.168.1.1) from the pool. This is the only address available.
PC2 (asks second)	Gets no address from the server. ipconfig shows IP Address 0.0.0.0, Subnet Mask 0.0.0.0 and Default Gateway 0.0.0.0, so the DHCP request failed.



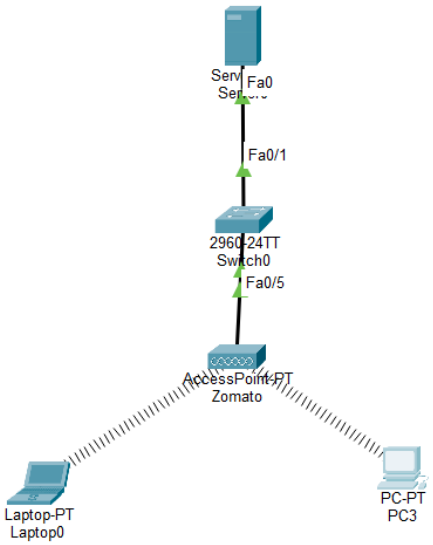
Screenshot: pool SingleIP_Pool with Maximum Number of Users = 1 (highlighted). PC1 received 192.168.1.100, and PC2 shows 0.0.0.0 (no IP address)

Why only one PC succeeds: The DHCP pool contains only one IP address. The first PC that sends a request receives that address and the server marks it as leased. When the second PC sends a Discover message, the server has no free address left in the pool, so it cannot send an Offer, and the request fails. The second PC will get an address only when the lease of the first one ends or the pool is made larger.

Task 4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Answer:

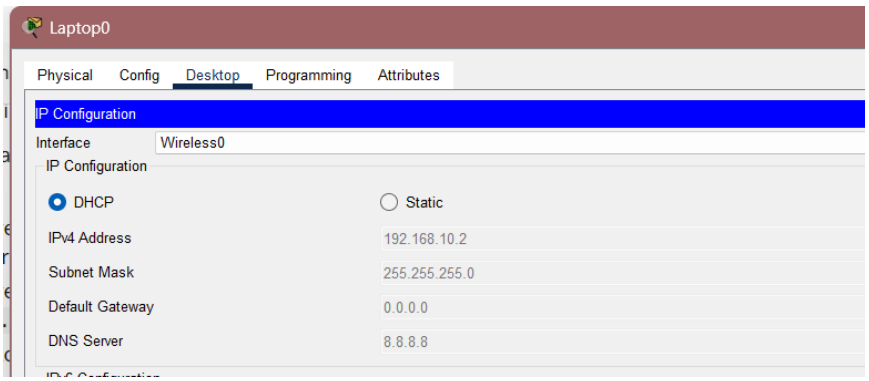
Devices: 1 Server-PT (Server0, the DHCP server), 1 switch (2960-24TT, Switch0), 1 Access Point (AccessPoint-PT named Zomato, the restaurant Wi-Fi), 1 Laptop-PT (Laptop0) and 1 PC-PT (PC3) used as the smartphone. The server is connected to the switch (Fa0 > Fa0/1), the switch is connected to the access point (Fa0/5), and the laptop and the smartphone connect to the access point wirelessly.



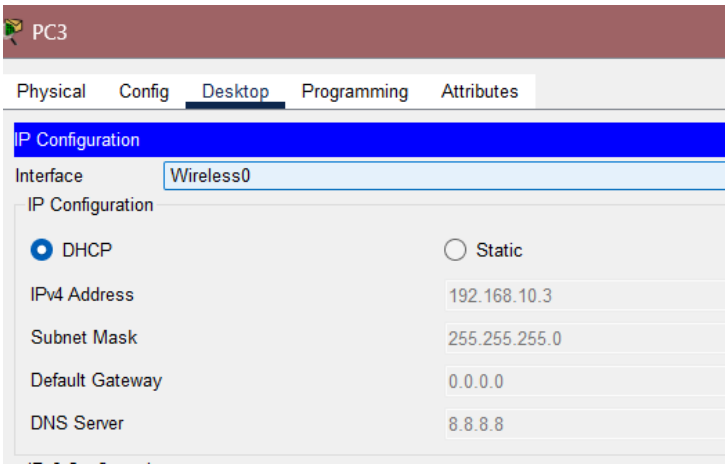
Screenshot: restaurant Wi-Fi topology (Server0, Switch0, access point "Zomato", Laptop0 and PC3)

1. Place the devices and connect Server0 to Switch0, and Switch0 to the Access Point (Zomato) with copper straight-through cables.
2. Server > Desktop > IP Configuration > Static: IP 192.168.10.1, mask 255.255.255.0.
3. Server > Services > DHCP: Service On. Make a pool with Start IP = 192.168.10.2, Mask = 255.255.255.0, DNS Server = 8.8.8.8, and save it.
4. On the Laptop and on PC3 (the smartphone): replace the wired module with the wireless module (WPC300N / WMP300N) so that they can connect to the Access Point. Their Wireless0 interface connects to the access point automatically (the SSID is the same).
5. On the Laptop and on PC3: Desktop > IP Configuration > Wireless0 > select DHCP.
6. Note the IP address received by each device.

Device	IP address assigned	Other values received
Laptop (Laptop0)	192.168.10.2	Mask 255.255.255.0, DNS 8.8.8.8, gateway 0.0.0.0 (not set in the pool)
Smartphone (PC3)	192.168.10.3	Mask 255.255.255.0, DNS 8.8.8.8, gateway 0.0.0.0 (not set in the pool)



Screenshot: Laptop0 > IP Configuration (Wireless0, DHCP) shows IP 192.168.10.2



Screenshot: PC3 (smartphone) > IP Configuration (Wireless0, DHCP) shows IP 192.168.10.3

Result: Both wireless devices got an IP address from the DHCP server automatically. The laptop received 192.168.10.2 and the smartphone received 192.168.10.3. They are different, in the same network 192.168.10.0/24, and the server address 192.168.10.1 was not given to any client. The default gateway shows 0.0.0.0 because no gateway was entered in the pool; if the restaurant had a router, its address would be entered as the default gateway in the pool.

Why DHCP is useful in this restaurant Wi-Fi situation:

1. Many customers come and go every day, and each one connects a phone or laptop. DHCP gives every device an IP address, mask, gateway and DNS automatically, so no one has to enter network settings by hand.
2. It avoids IP address conflicts, because the server keeps a record and never gives the same address to two devices at the same time.
3. The addresses are leased, so when a customer leaves, the address returns to the pool and can be given to the next customer. A small pool can serve many visitors over time.
4. The staff only manage the settings in one place (the DHCP server), and customers can quickly open apps like Zomato to order or track food without any setup.